

# WASP-36b

R-2371

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-36\_130224 · created 2026-08-20T00:26:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456347.74418 +/- 0.00051 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.16 +/- 0.046                    |
| Transit depth (Rp/Rs)^2                           | 2.55 +/- 1.49 [%]                 |
| Orbital Inclination (inc)                         | 81.39 +/- 2.73                    |
| Airmass coefficient 1 (a1)                        | 1.094 +/- 0.033                   |
| Airmass coefficient 2 (a2)                        | -0.067 +/- 0.095                  |
| Scatter in the residuals of the lightcurve fit is | 3.01 %                            |
| Best Comparison Star                              | #3 - [607, 85]                    |
| Optimal Method                                    | PSF photometry                    |
| Transit Duration (day)                            | 0.056 +/- 0.029                   |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                         |
|---------------------------|-----------------------------------------|
| Rp/R* fit vs published    | 0.16 ± 0.046 vs 0.1368 (deviation 0.5σ) |
| Duration fit vs published | 0.056 d vs 0.0758 d (deviation 0.68σ)   |
| Mid-time O–C              | -1.1 min                                |
| Depth SNR                 | 3.5                                     |
| Residual scatter          | 3.01 %                                  |

## Light curve

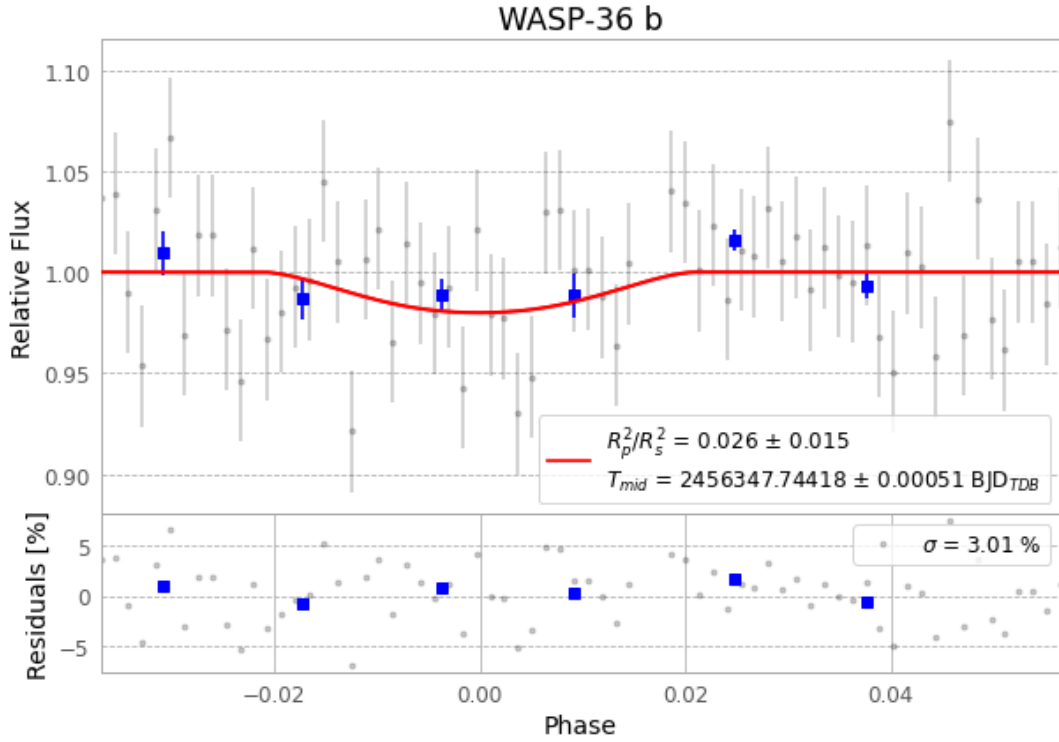


Figure 1 — FinalLightCurve\_WASP-36 b\_2013-02-23.png (SHA-256 ae74736703cecd8...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 311], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 39d714a0af411dda...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

70 FITS frames (46 MB), WASP-36130224042112.FITS → WASP-36130224074812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-130224.log (424edf4fcdcc...), FinalLightCurve\_WASP-36 b\_2013-02-23.png (ae74736703ce...), FinalLightCurve\_WASP-36 b\_2013-02-23.csv (25470740a4fa...)

## Compute resources

Wall time 487.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:26Z.